SUMMARY OF Project – 3 DAX Depo

- First, I carefully read the complete project instructions and understood that this project is mainly focused on **data modeling and advanced DAX calculations**, not on creating dashboards or charts.
- I noted clearly that **only Matrix visual is allowed** and all results must be shown through DAX measures.
- I downloaded all the given datasets and imported them into Power BI.
- The tables used in the project are **Sales_Fact_Table, Returns_Fact_Table, Customer_Dimension, Product_Dimension, Date_Dimension, and Region_Dimension**.
- After importing the data, I opened **Model View** and identified the **fact tables** and **dimension tables**.
- **Fact tables:** Sales_Fact_Table and Returns_Fact_Table
- **Dimension tables:** Customer_Dimension, Product_Dimension, Date_Dimension, Region_Dimension
- I checked the columns in each table and identified the **primary keys and foreign keys** required for relationships.
- Wherever needed, I cleaned column names and ensured correct data types for proper relationship creation.
- I then created **one-to-many relationships** from all dimension tables to the Sales_Fact_Table using appropriate keys such as CustomerID, ProductID, DateKey, and RegionID.
- I also connected the **Returns_Fact_Table to the Sales_Fact_Table** using the SalesID to track returned transactions.
- All relationships were kept **single-directional**, and a proper **star schema** was maintained.
- After completing the data model, I created required **calculated columns** such as Profit, ReturnFlag, and Customer Full Name using DAX as instructed.
- To organize calculations properly, I created a **separate dedicated Measure table**.
- All DAX measures were created inside this table to keep the model clean and structured.
- I then created all required **measures**, including:
- Total Sales

- Total Cost
- Total Profit
- Return Rate
- Average Sales
- YTD Sales, YoY calculations, and Running Total
- I also used iterator functions like **SUMX and AVERAGEX**, conditional logic using **IF and SWITCH**, and filter context functions like **ALL, FILTER, and CALCULATE** as required.
- For time-based analysis, I used **TOTALYTD, SAMEPERIODLASTYEAR, and DATESBETWEEN** functions based on the Date_Dimension table.
- Finally, I displayed all calculated results **only using Matrix visuals**, grouped by Region, Month, Product Category, and Customer details, as mentioned in the project instructions.